Fiscal Year:	FY 2014	**Task Last Updated:**	FY 09/08/2013
PI Name:	O'Banion, Kerry M.D., Ph.D.		
Project Title:	Impact of Space Radiation Induced Neuroinflammation on Alzheimer and Parkinson Disease Pathology		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Rochester	**State:**	NY
Zip Code:	14642-0001	**Congressional District:**	25
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2011 Space Radiobiology NNJ11ZSA001N
Start Date:	11/13/2012	**End Date:**	11/12/2015
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Simonsen, Lisa	**Contact Phone:**	
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	September 2013: Sean Hurley was listed in this record as a Co-Investigator; however, Sean left the University prior to the start of this grant and was probably inadvertently included because of the delay between the initial application and the actual funding date. I removed his name from Section 7.		
COI Name (Institution):	Olschowka, John (University of Rochester Medical Center) Williams, Jacqueline (University of Rochester Medical Center)		
Grant/Contract No.:	NNX13AC33G		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The brain reacts to injury by activation of glial cells and expression of inflammatory mediators in a process called neuroinflammation. Neuroinflammation occurs in the setting of age-associated neurodegenerative diseases such as Alzheimer's and Parkinson's and numerous studies suggest a contributory role to disease pathogenesis. We have previously shown that space radiation, at doses approaching those that will be experienced by astronauts on long missions to the moon and Mars, elicits brain neuroinflammation. Therefore we propose to investigate the possibility that space radiation influences pathological changes associated with neurodegenerative diseases. Our studies will utilize three different mouse models of the two most prevalent age-associated neurodegenerative diseases (Alzheimer's and Parkinson's) to assess the effects of space radiation on disease-associated pathology and behavioral changes as well as indices of neuroinflammation and oxidative stress. Findings from these studies will help to refine risk estimates for future space explorers and may offer possible strategies for mitigation of such risks.		
Rationale for HRP Directed Research:			
Research Impact/Earth Benefits:			
Task Progress:	In this first year of the grant we published results arising from an initial investigation of space radiation effects on Alzheimer's pathogenesis in an APP/PS-1 double transgenic mouse model that shows amyloid beta plaque accumulation starting at 4 to 5 months of age. These studies were carried out as part of my previous grant from NASA (NNX08BA09G) and were reported in PLoS One (Cherry et al., 2012. PloS One 7:e53275). The main findings of that study were that male APP/PS-1 transgenic mice treated with 100 cGy 56Fe particles (1 GeV/μ) at 3.5 months of age showed deficits in two behavioral tasks and increased plaque deposition when examined 6 months later. Behavioral deficits were also seen with 10 cGy 56Fe and the amyloid deposition trended toward significance. Although female mice were included in the study, early death of this cohort led us to sacrifice female mice at an earlier age and while some behavioral deficits were apparent, there was no evidence for increased plaque deposition. Unfortunately this age difference prevents us from making a clear conclusion about the differences observed between males and females. Mechanistically, we did not observe differences in microglial responses in irradiated vs. control transgenic male mice, but we did find increased ICAM-1 staining, suggesting a persistent activation of vascular endothelial cells. With regard to the current grant, we are making progress on both Specific Aims 1 and 2. In particular, we have arranged production through Jackson Laboratories of all transgenic mice required for these two aims. Thus, we have 140 triple transgenic mice (half male, half female) currently aging at Jackson Laboratories that will be shipped to Brookhaven National Laboratories in October for use in three separate irradiation campaigns during NSRL-13C. The specific dates and beams are: 1) 300 MeV/μ Silicon at 10 and 100 cGy on October 31; 2) 600 MeV/μ Iron at 10 and 100 cGy on November 15; and 3) 200 cGy mixed protons mimicking a solar flare on November 14. For each ion and energy 10 male and 10 female mice will be irradiated. Two set of sham irradiated (control) mice with 10 males and 10 females in each set will be included in the experiment. Mice will be 7-8 months of age at the time of irradiation and will be shipped back to Rochester to be aged for an additional year before behavioral, histological, and molecular analyses as described in Aims 1 and 3. Similarly, we have arranged with Jackson Laboratories to have Parkin-Q311X mice derived for use in Aim 2. Specifically, we contracted with Jackson to do a rapid derivation of mice with a goal of 80 mice (40 male, 40 female) harboring the disease model transgene and 80 control littermates. Importantly, this provides a non-transgenic control group that will allow us to know whether behavioral effects of radiation interact with genotype. The breeding is on target for having mice at the right age (about 6 months) at the time of irradiation, which we have requested be scheduled during the Spring 2014 NSRL run (NSRL14A).		
Bibliography Type:	Description: (Last Updated: 01/07/2026)		

Abstracts for Journals and Proceedings	O'Banion MK, Cherry JD, Lemere CA, Williams JP, Olschowka JA. "Galactic cosmic radiation leads to cognitive impairment and increased Aß accumulation in a mouse model of Alzheimer's disease." 2013 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 12-14, 2013. 2013 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 12-14, 2013. , Feb-2013
Abstracts for Journals and Proceedings	O'Banion MK, Cherry JD, Lemere CA, Williams JP, Olschowka JA. "HZE particle irradiation leads to cognitive impairment and increased Aß accumulation in a mouse model of Alzheimer's disease." 24th Annual NASA Space Radiation Investigators' Workshop. Part of HITSRS2013--Heavy Ion in Therapy and Space Radiation Symposium 2013, Chiba, Japan, May 15-18, 2013. 24th Annual NASA Space Radiation Investigators' Workshop. Part of HITSRS2013--Heavy Ion in Therapy and Space Radiation Symposium 2013, Chiba, Japan, May 15-18, 2013. , May-2013
Abstracts for Journals and Proceedings	O'Banion MK, Cherry JD, Lemere CA, Liu B, Frost JL, Williams JP, Olschowka JA. "Space radiation exacerbates amyloid pathology and cognitive dysfunction in the APP/PS---1 transgenic mouse model of Alzheimer's disease." 11th International Conference on Alzheimer's and Parkinson's Diseases, Florence, Italy, March 06--10, 2013. Neurodegenerative Diseases 2013;Suppl 1:84. 11th International Conference on Alzheimer's and Parkinson's Diseases, Florence, Italy, March 06--10, 2013. , Mar-2013
Abstracts for Journals and Proceedings	O'Banion MK. "Galactic Cosmic Radiation Exacerbates Cognitive Impairment and Amyloid Plaque Accumulation in a Mouse Model of Alzheimer's Disease." International Symposium on Radiation Science, Taipei, Taiwan, May 20-21, 2013. Invited speaker. International Symposium on Radiation Science, Taipei, Taiwan, May 20-21, 2013. , May-2013
Abstracts for Journals and Proceedings	O'Banion MK, Cherry JD, Lemere CA, Williams JP, Olschowka JA. (O'Banion MK=Invited speaker) "HZE Irradiation Enhances Amyloid Deposition and Cognitive Deficits in a Mouse Model of Alzheimer's Disease." 59th Annual Meeting of the Radiation Research Society, New Orleans, LA, September 14-18, 2013. Symposium on Radiation--induced non---malignant diseases: cataract, cardiovascular disease and neurocognitive effects, and interactions with metabolic disease. O'Banion MK=Invited speaker. Proceedings of 59th Annual Meeting of the Radiation Research Society, New Orleans, LA, September 14-18, 2013. Presentation number: S2505. http://www.abstractsonline.com/Plan/ViewAbstract.aspx?sKey=a55b7035-4f8a-4d01-87b6-51a2c57a6d0c&cKey=db07fd40-9f10-40cb-a9e5-b6b923ff6162&mKey=01c994ce-9545-4ec7-8580-d74360fa8373 ; accessed 11/7/13. , Sep-2013
Articles in Peer-reviewed Journals	Cherry JD, Liu B, Frost JL, Lemere CA, Williams JP, Olschowka JA, O'Banion MK. "Galactic cosmic radiation leads to cognitive impairment and increased aß plaque accumulation in a mouse model of Alzheimer's disease." PLoS One. 2012;7(12):e53275. http://dx.doi.org/10.1371/journal.pone.0053275 ; PubMed PMID: 23300905 , Dec-2012